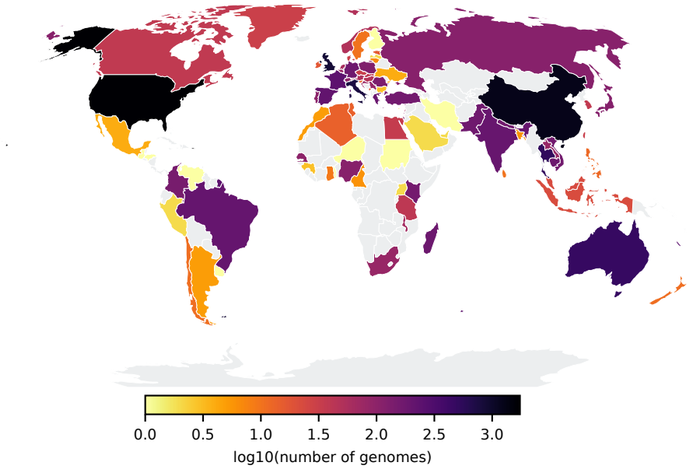
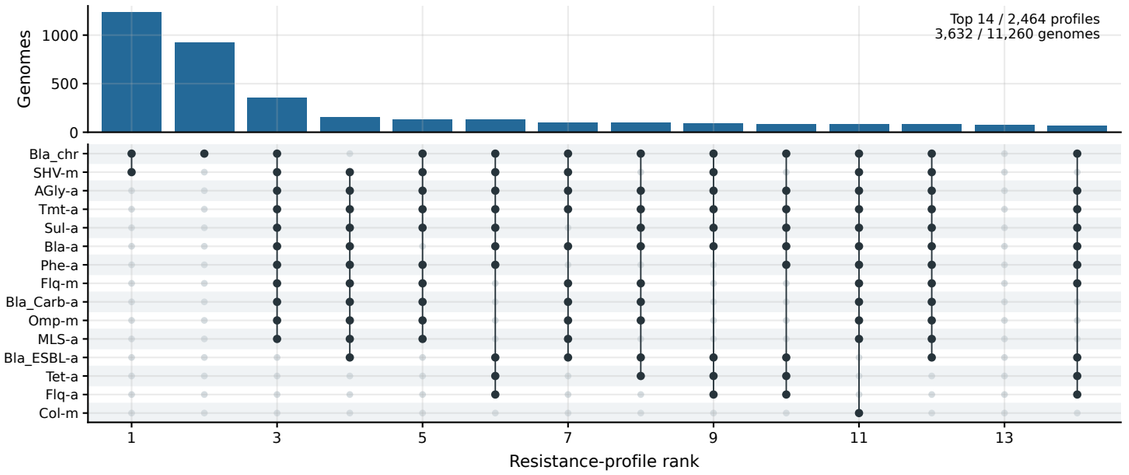
## A Global sampling




## B Resistance-state diversity




## C Generalization to held-out lineages




## D Shared resistance-acquisition architecture




## E Lineage-dependent empirical acquisition shifts




## F Country-associated acquisition heterogeneity




Africa Americas Asia Europe Oceania No reconstructed gains